A First Course in Abstract Algebra

Section 1: Introduction and Examples

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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1 Introduction and Examples

Exercise 1.1: A cube of i

$i^3 = i^2 i = -i$, so the answer is $0 - i$.

Exercise 1.2: A fourth power of i

$i^4 = (i^2)^2 = (-1)^2 = 1$, or $1 + 0i$.

Exercise 1.3: A high power of i

Since $23 = 4 \cdot 5 + 3$ and $i^4 = 1$, $i^{23} = (i^4)^5 i^3 = -i$.

Exercise 1.4: A power of minus i

$(-i)^{35} = (-1)^{35} i^{35} = -i^3 = i$, because $35 \equiv 3 \pmod 4$.

Exercise 1.5: Multiplying two binomials

Distribute and use $i^2 = -1$: $(4 - i)(5 + 3i) = 20 + 12i - 5i - 3i^2 = 23 + 7i$.

Exercise 1.6: Another binomial product

$(8 + 2i)(3 - i) = 24 - 8i + 6i - 2i^2 = 26 - 2i$. The real contribution from $-2i^2$ is $+2$.

Exercise 1.7: A product plus a complex number

$(2 - 3i)(4 + i) = 8 + 2i - 12i - 3i^2 = 11 - 10i$. Adding $6 - 5i$ yields $17 - 15i$.

Exercise 1.8: Cubing one plus i

First $(1 + i)^2 = 1 + 2i + i^2 = 2i$. Thus $(1 + i)^3 = 2i(1 + i) = -2 + 2i$.

Exercise 1.9: A fifth power by the binomial theorem

Expand $(1 - i)^5 = 1 - 5i + 10i^2 - 10i^3 + 5i^4 - i^5$. Replacing $i^2 = -1, i^3 = -i, i^4 = 1, i^5 = i$ gives $(1 - 10 + 5) + (-5 + 10 - 1)i = -4 + 4i$.

Exercise 1.10: The modulus of three minus four i

For $a + bi$ the modulus is $\sqrt{a^2 + b^2}$. Hence $|3 - 4i| = \sqrt{9 + 16} = 5$.

Exercise 1.11: The modulus of six plus four i

$|6 + 4i| = \sqrt{36 + 16} = \sqrt{52} = 2\sqrt{13}$. The modulus is nonnegative.

Exercise 1.12: Polar form: Three minus four i

Compute $r = |3 - 4i| = 5$ and divide the real and imaginary coordinates by r. This gives

$$3 - 4i = 5\left(3/5 + (-4/5)\, i\right).$$

The two displayed normalized coordinates have squared sum 1, so the parenthesized complex number lies on the unit circle. This is the requested form $r(p + qi)$.

Exercise 1.13: Polar form: Minus one plus i

Compute $r = |-1 + i| = \sqrt{2}$ and divide the real and imaginary coordinates by r. This gives

$$-1 + i = \sqrt{2}\left(-1/\sqrt{2} + \left(1/\sqrt{2}\right) i\right).$$

The two displayed normalized coordinates have squared sum 1, so the parenthesized complex number lies on the unit circle. This is the requested form $r(p + qi)$.

Exercise 1.14: Polar form: Twelve plus five i

Compute $r = |12 + 5i| = 13$ and divide the real and imaginary coordinates by r. This gives

$$12 + 5i = 13\left(12/13 + (5/13)\, i\right).$$

The two displayed normalized coordinates have squared sum 1, so the parenthesized complex number lies on the unit circle. This is the requested form $r(p + qi)$.

Exercise 1.15: Polar form: Minus three plus five i

Compute $r = |-3 + 5i| = \sqrt{34}$ and divide the real and imaginary coordinates by r. This gives

$$-3 + 5i = \sqrt{34}\left(-3/\sqrt{34} + \left(5/\sqrt{34}\right) i\right).$$

The two displayed normalized coordinates have squared sum 1, so the parenthesized complex number lies on the unit circle. This is the requested form $r(p + qi)$.

Exercise 1.16: Fourth roots of one

Write $z = re^{i\theta}$. Then $r^4 = 1$, so $r = 1$, and $4\theta = 2\pi k$. Distinct roots occur for $k = 0, 1, 2, 3$:

$$z = e^{i\pi k/2} \in \{1, i, -1, -i\}.$$

Larger integers repeat these roots; a fourth-degree polynomial has no more than four distinct roots.

Exercise 1.17: Fourth roots of minus one

Since $-1 = e^{i(\pi+2\pi k)}$, the roots are

$$z = e^{i(\pi/4+k\pi/2)}, \quad k = 0, 1, 2, 3.$$

In Cartesian form these are $(1+i)/\sqrt{2}$, $(-1+i)/\sqrt{2}$, $(-1-i)/\sqrt{2}$, and $(1-i)/\sqrt{2}$. Each has modulus 1 and fourth power -1.

Exercise 1.18: Cube roots of minus eight

Here $-8 = 8e^{i(\pi+2\pi k)}$. Thus $r = 2$ and $\theta = \pi/3 + 2\pi k/3$. For $k = 0, 1, 2$ the roots are $1 + i\sqrt{3}$, -2, and $1 - i\sqrt{3}$. Distinct arguments modulo 2π give all three solutions.

Exercise 1.19: Cube roots of minus twenty-seven i

Use $-27i = 27e^{i(-\pi/2+2\pi k)}$. The roots are $3e^{i(-\pi/6+2\pi k/3)}$ for $k = 0, 1, 2$, namely

$$\frac{3\sqrt{3}}{2} - \frac{3}{2}i, \qquad 3i, \qquad -\frac{3\sqrt{3}}{2} - \frac{3}{2}i.$$

Cubing triples each argument and cubes the modulus, giving $-27i$ in every case.

Exercise 1.20: Sixth roots of one

The six roots are $e^{i\pi k/3}$, $k = 0, \ldots, 5$. Explicitly:

$$1, \quad \tfrac{1}{2} + \tfrac{\sqrt{3}}{2}i, \quad -\tfrac{1}{2} + \tfrac{\sqrt{3}}{2}i, \quad -1, \quad -\tfrac{1}{2} - \tfrac{\sqrt{3}}{2}i, \quad \tfrac{1}{2} - \tfrac{\sqrt{3}}{2}i.$$

They follow from $r^6 = 1$ and $6\theta \in 2\pi\mathbb{Z}$.

Exercise 1.21: Sixth roots of minus sixty-four

Here $r^6 = 64$ gives $r = 2$ and $6\theta = \pi + 2\pi k$. Thus $z = 2e^{i(\pi/6+k\pi/3)}$, $k = 0, \ldots, 5$, or

$$\sqrt{3} + i, \quad 2i, \quad -\sqrt{3} + i, \quad -\sqrt{3} - i, \quad -2i, \quad \sqrt{3} - i.$$

These have six distinct arguments modulo 2π.

Exercise 1.22: Addition modulo seventeen

$10 + 16 = 26 = 17 + 9$, so $10 +_{17} 16 = 9$. In modular addition, choose the unique representative between zero inclusive and the modulus exclusive.

Exercise 1.23: Addition modulo ten

$8 + 6 = 14 = 10 + 4$, so $8 +_{10} 6 = 4$. In modular addition, choose the unique representative between zero inclusive and the modulus exclusive.

Exercise 1.24: Real modular addition

$20.5 + 19.3 = 39.8 = 25 + 14.8$, so the representative in $[0, 25)$ is 14.8. In modular addition, choose the unique representative between zero inclusive and the modulus exclusive.

Exercise 1.25: Addition modulo one

$1/2 + 7/8 = 11/8 = 1 + 3/8$, hence $1/2 +_1 7/8 = 3/8$. In modular addition, choose the unique representative between zero inclusive and the modulus exclusive.

Exercise 1.26: Angles modulo two pi

$3\pi/4 + 3\pi/2 = 9\pi/4 = 2\pi + \pi/4$, so the result is $\pi/4$. In modular addition, choose the unique representative between zero inclusive and the modulus exclusive.

Exercise 1.27: An irrational modulus

$2\sqrt{2} + 3\sqrt{2} = 5\sqrt{2}$, and $\sqrt{32} = 4\sqrt{2}$. Subtracting that modulus once leaves $\sqrt{2} \in [0, 4\sqrt{2})$, the answer. In modular addition, choose the unique representative between zero inclusive and the modulus exclusive.

Exercise 1.28: Inputs outside the underlying set

The operation $+_6$ on $\mathbb{R}_6 = [0, 6)$ accepts only ordered pairs of elements of that set. The number 8 is not an element of $\mathbb{R}_6$, so $5 +_6 8$ is not an operation on the stated domain. One could first replace 8 by 2, but that is an additional reduction, not the given expression as written.

Exercise 1.29: A linear equation modulo fifteen

The congruence is $x + 7 \equiv 3 \pmod{15}$, so $x \equiv -4 \equiv 11 \pmod{15}$. The only representative in $\mathbb{Z}_{15}$ is 11, and $11 +_{15} 7 = 3$.

Exercise 1.30: An angular equation

Subtract $3\pi/2$ from $3\pi/4$: $x \equiv -3\pi/4 \pmod{2\pi}$. Adding 2π gives $x = 5\pi/4 \in [0, 2\pi)$. Its sum with $3\pi/2$ is $11\pi/4$, whose remainder is $3\pi/4$.

Exercise 1.31: Twice a residue modulo seven

We need $2x \equiv 3 \pmod{7}$. Since $2 \cdot 4 \equiv 1$, multiplication by 4 gives $x \equiv 12 \equiv 5$. Checking, $5 +_7 5 = 3$. The inverse of 2 makes the answer unique.

Exercise 1.32: Three times a residue modulo seven

The equation is $3x \equiv 5 \pmod 7$. Since $3 \cdot 5 \equiv 1$, we get $x \equiv 25 \equiv 4$. Indeed $4 + 4 + 4 = 12 \equiv 5 \pmod 7$.

Exercise 1.33: Twice a residue modulo twelve

The equation $2x \equiv 2 \pmod{12}$ means $12 \mid 2(x-1)$, equivalently $6 \mid x - 1$. Representatives in $\mathbb{Z}_{12}$ are therefore $x = 1$ and $x = 7$. Both give remainder 2 when doubled. Dividing a congruence here also divides its modulus.

Exercise 1.34: Four times a residue modulo four

Every residue $x \in \{0, 1, 2, 3\}$ satisfies $4x \equiv 0 \pmod 4$. Hence all four residues are solutions; the coefficient has no multiplicative inverse modulo 4.

Exercise 1.35: An isomorphism from eighth roots

Multiplication of powers corresponds to addition, so the image of ζ^m must be $5m \pmod 8$. For the six requested exponents:

m	0	3	4	5	6	7
$5m \bmod 8$	0	7	4	1	6	3

This also gives $m = 2 \mapsto 2$, as required in the question. Since $\gcd(5, 8) = 1$, all eight images are distinct.

Exercise 1.36: An isomorphism from seventh roots

The image of ζ^m is $4m \pmod 7$ because powers become repeated sums. Thus

m	0	2	3	4	5	6
$4m \bmod 7$	0	1	5	2	6	3

Multiplication by 4 is a permutation of $\mathbb{Z}_7$, since 4 is relatively prime to 7.

Exercise 1.37: Why the proposed image is impossible

The element $\zeta = e^{i\pi/3}$ generates all six roots, so its image under an isomorphism must generate $\mathbb{Z}_6$. But 4 generates only $\{0, 4, 2\}$, since $3 \cdot 4 \equiv 0 \pmod 6$. Alternatively, $\zeta^3 = -1 \neq 1$ would map to 0, violating injectivity. Thus no isomorphism can have this image.

Exercise 1.38: Angle addition from Euler formula

Compute the product in two ways:

$$e^{ia}e^{ib} = e^{i(a+b)} = \cos(a + b) + i \sin(a + b),$$

whereas direct multiplication gives

$$(\cos a + i\sin a)(\cos b + i\sin b) = (\cos a\cos b - \sin a\sin b) + i(\sin a\cos b + \cos a\sin b).$$

Equality of real parts proves the cosine identity; equality of imaginary parts proves the sine identity. This argument explains why multiplying unit complex numbers adds their angles.

Exercise 1.39: Multiplying polar forms

Put $r_j = |z_j|$. Distributing the factors gives

$$z_1 z_2 = r_1 r_2\{(\cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2) + i(\sin\theta_1\cos\theta_2 + \cos\theta_1\sin\theta_2)\}.$$

The two addition identities replace the braces by $\cos(\theta_1+\theta_2) + i\sin(\theta_1+\theta_2)$. Thus the moduli multiply and the angles add. If one number is zero, the same product formula is zero regardless of the chosen angle.

Exercise 1.40: The triple-angle identity

(a) Cubing Euler's formula gives $e^{3i\theta} = (\cos\theta + i\sin\theta)^3$. Its real part is $\cos^3\theta - 3\cos\theta\sin^2\theta$, since the terms with odd powers of i are imaginary and $i^2 = -1$. Hence $\cos 3\theta = \cos^3\theta - 3\cos\theta\sin^2\theta$.

(b) Substitute $\sin^2\theta = 1 - \cos^2\theta$ to obtain $\cos 3\theta = \cos^3\theta - 3\cos\theta(1-\cos^2\theta) = 4\cos^3\theta - 3\cos\theta$.

Exercise 1.41: Euler's formula from power series

Start with $e^z = \sum_{n=0}^{\infty} z^n/n!$ and substitute $z = i\theta$. Separate even and odd powers using $i^{2k} = (-1)^k$ and $i^{2k+1} = i(-1)^k$:

$$\begin{aligned} e^{i\theta} &= \sum_{k=0}^{\infty} \frac{(-1)^k\theta^{2k}}{(2k)!} + i\sum_{k=0}^{\infty}\frac{(-1)^k\theta^{2k+1}}{(2k+1)!} \\ &= \cos\theta + i\sin\theta. \end{aligned}$$

The two real series are exactly the given cosine and sine expansions. This is the requested formal derivation; their absolute convergence for every real θ also justifies separating the two subseries analytically.